

In-frame deletion of lasB in LB056 via pEXG2 allelic exchange

Primer design • PCR / Assembly protocol • 2026-05-07

1 Konstrukt-Überblick

Zielgen	lasB in LB056 (1497 nt, 498 aa)
Strategie	In-frame Deletion, Scar 20 aa (15 N + 5 C)
Vektor	pEXG2 (5084 bp), HindIII-linearisiert
Methode	In-Fusion 3-Fragment (Vektor + UP + DN)
Counterselektion	sacB / GmR (pEXG2)
Finales Plasmid	pEXG2-ΔlasB_LB056 • 6152 bp

2 Primer

Primer	Sequenz 5'→ 3' (Tail fett, Body normal)	nt	Tm _{body}	GC
P1_lasB_UP_Fwd	CATAAATGTAAAGCATGGTGCAGCAGCGGATCG	33	63.1 °C	67 %
P2_lasB_UP_Rev	CAACGCGCTCGGGCACATGATCGCAACGAACAACAGGTCAAG	42	63.0 °C	48 %
P3_lasB_DN_Fwd	TTCGTTGCGATCATGTGCCCGAGCGCGTTGTAAG	34	63.2 °C	63 %
P4_lasB_DN_Rev	CGACCTGCAGAAGCTCGCCAGGTACTCGCCTTGC	34	63.2 °C	68 %

3 PCR-Bedingungen (B7 High Fidelity)

Tm _{body} Mittel	63.1 °C (Spread 0.28 °C)
Annealing Ta	63 °C (= Tm _{body} ; Gradient ± 5 °C empfohlen falls 1. Versuch fehlschlägt)
Elongation	60 s/kb • UP 579 bp → 35 s • DN 549 bp → 33 s

4 Konstrukt-Zusammenfassung

UP-Amplicon 579 bp • DN-Amplicon 549 bp • Insert 1098 bp • Final 6152 bp

Scar-Sequenz (15 N + 5 C aa + Stop):

ATG AAG AAG GTT TCT ACG CTT GAC CTG TTG TTC GTT GCG ATC ATG | TGC CCG AGC GCG TTG TAA
M K K V S T L D L L F V A I M | C P S A L *

Off-Target Scan: PASS • HindIII-Sites in finalem Plasmid: 0 (erwartet 0)